VPN troubleshooting guide for remote workers

Article KB0010016 | Owner: Network Support | Subcategory: VPN | Applies to: Windows 11 23H2/24H2 (Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840), macOS 14/15 (MacBook Pro/Air), Palo Alto GlobalProtect VPN client, Remote workers connecting from home or other non-Acme networks

Summary

This guide helps remote workers diagnose and resolve common issues with the Acme Global VPN (Palo Alto GlobalProtect). It covers symptoms such as connection failures, frequent drops, authentication errors, certificate problems, and inability to reach internal resources. Users can follow a self-help procedure to renew certificates, update the client, flush DNS, and adjust split-tunnel settings, while service-desk agents have a structured workflow for deeper troubleshooting and escalation.

Symptoms

- "Unable to connect to VPN" or "Connection timed out" when launching GlobalProtect
- Frequent disconnections after a few minutes of use
- Authentication failed - message: "Certificate validation failed" or "Invalid credentials"
- VPN connects but internal sites (e.g., intranet.acme.example, file shares) are unreachable
- Error: "VPN certificate has expired" shown in the client UI

Common causes

- Expired or corrupted machine certificate used for GlobalProtect authentication
- Outdated GlobalProtect client version lacking compatibility with current portal settings
- DNS cache stale or incorrect split-tunnel configuration preventing access to internal subnets
- Home router or ISP blocking UDP ports 500/4500 or TCP 443 used by GlobalProtect
- Incorrect time/date on the endpoint causing certificate validation to fail

Preliminary Checks

Before diving into troubleshooting, verify a few basic items that often resolve VPN issues quickly. Ensure the device is connected to the internet (open a public website). Confirm the system clock is set correctly; an incorrect time will cause certificate validation to fail. Verify that the GlobalProtect icon shows a status of "Disconnected" or "Connecting..." and note any error message displayed.

Self-Help Procedure (User)

Follow these steps to resolve the most common VPN problems. If the issue persists after completing the procedure, gather the information noted in each step and contact the IT Service Desk via phone ext. 4357, the Teams "IT Help" channel, or the self-service portal https://it.acme.example.

1. 1. **Check and renew the VPN certificate**: Open the GlobalProtect client, go to Settings -> Authentication. If the certificate status shows "Expired" or "Not Valid", click "Renew Certificate" or "Re-issue Certificate". This triggers a request to the Network Support team; you will receive a notification when the new certificate is ready. Import it when prompted.
2. **Update the GlobalProtect client**: In the client, select Check for Updates. If an update is available, install it and restart the client. The latest version is always compatible with the portal vpn.acme.example.
3. **Flush DNS and reset split-tunnel**: Open a command prompt (Windows) or Terminal (macOS) and run `ipconfig /flushdns` (Windows) or `sudo dscacheutil -flushcache; sudo killall -HUP mDNSResponder` (macOS). Then, in GlobalProtect Settings -> Network, ensure "Split Tunnel" is enabled and that the list includes the Acme internal subnets (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16). Apply changes and reconnect.
4. **Test alternative network**: Disconnect from your home Wi-Fi and connect via a mobile hotspot. Attempt to connect to VPN again. If it succeeds, the home router is likely blocking the required ports. Log into the router and open UDP 500, UDP 4500, and TCP 443, or enable VPN passthrough. Save settings and reconnect to your home Wi-Fi.
5. **Verify connection and internal access**: Once connected, try to reach an internal site such as https://intranet.acme.example or a file share (`\fileserver.acme.example/share`). If the page loads or the share opens, the VPN is working correctly.

Service Desk Procedure (Agent)

When a user reports a VPN issue after attempting self-help, follow this structured workflow. Document each step in the ticket and communicate progress to the user.

1. 1. **Validate certificate status**: Ask the user to confirm the certificate expiration date shown in GlobalProtect -> Settings -> Authentication. If expired, initiate a certificate re-issue request via the Network Support queue (provide user name, employee ID, and device hostname). Notify the user once the new certificate is available and guide them to import it.
2. **Confirm client version**: Have the user check GlobalProtect -> About. Compare against the approved version listed in the Network Support Confluence page. If outdated, push the latest installer through Intune or provide a direct download link from the internal software portal.
3. **Review DNS and split-tunnel settings**: Request the user to run `ipconfig /displaydns` (Windows) or `scutil --dns` (macOS) and share the output. Verify that internal DNS suffixes (acme.example) are present and that split-tunnel includes the required subnets. If mis-configured, push a corrected configuration profile via Intune or guide the user to adjust settings manually.
4. **Check for network-level blocks**: Ask the user to test connectivity using a mobile hotspot. If VPN works on hotspot, instruct the user to enable UDP 500/4500 and TCP 443 passthrough on their home router or contact their ISP. Provide a generic router-port-forwarding guide if needed.
5. **Escalate if needed**: If the VPN still fails after certificate, client, DNS, and network checks, collect logs: GlobalProtect logs (`%ProgramData%/Palo Alto Networks/GlobalProtect/logs/` on Windows, `/opt/paloaltonetworks/globalprotect/logs/` on macOS) and a screenshot of the error. Escalate to the Network Support team (email: network-support@acme.example) with the ticket number, user details, and collected logs.

Verification

After applying any fix, confirm that the VPN is stable and that internal resources are reachable. Ask the user to perform the following checks and report results back to the service desk.

Additional Notes

Remember that VPN certificates are renewed yearly; users will receive a reminder email 30 days before expiry. Keeping the GlobalProtect client up to date is enforced via Intune compliance policies. For persistent issues unrelated to the client (e.g., ISP-level blocking), the Network Support team may work with the user's ISP or recommend a temporary shift to a corporate-provided LTE hotspot.

When to escalate

Escalate to the Network Support group when: the VPN certificate cannot be renewed through the self-service portal, the client fails to launch or crashes repeatedly, split-tunnel or DNS corrections do not restore access to internal sites, or the user confirms that the problem persists across multiple networks (home Wi-Fi, hotspot, and a different ISP). Include in the escalation ticket: user name, employee ID, device hostname and OS, GlobalProtect version, certificate expiry date, exact error messages, steps already taken, and any relevant logs or screenshots.

Keywords: VPN, GlobalProtect, certificate expired, authentication failed, cannot connect, VPN disconnects, split tunnel, DNS flush, home router block, hotspot test, Network Support, vpn.acme.example, Palo Alto, Intune, certificate renewal